AEGIS ALPHA LAB

A Mathematically Specified, Barra-Aware, Long-Short Equity Research Platform

V1 Specification with V2 / V3 Roadmap — Paper-Strengthened Edition

Version 3.0

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Methodological grounding

Huang & Fan (2026), "Beyond Prompting: An Autonomous Framework for Systematic Factor Investing via Agentic AI."

Wangpratham (2026), "BrainAlpha: An Autonomous Multi-Agent System for Quantitative Alpha Discovery."

McLean & Pontiff (2016); Harvey, Liu & Zhu (2016); López de Prado (2018).

1. Executive Summary

Aegis Alpha Lab is a long-short U.S. equity research platform built around three commitments: (i) every promoted signal must carry residual predictive power after standard style and industry risk premia are removed; (ii) every promotion decision must pass a pre-committed multiple-testing regime; and (iii) every stage of the pipeline must be reproducible from code and logs alone.

This edition of the specification tightens the framework against two recent papers that map directly onto our problem. Huang and Fan (2026) formalize the closed-loop agentic factor-investing workflow with explicit temporal isolation and σ-algebra-level no-lookahead constraints; we adopt that formalism verbatim. Wangpratham (2026) develops a structured 660-cell exploration grid, a typed failure-mode taxonomy, and a convergence-governed repair engine for LLM-based alpha discovery; we adopt that architecture as the target specification for V2.

What V1 delivers

- Point-in-time U.S. daily equity panel on CRSP + Compustat Point-in-Time via WRDS, covering ~4,000 common stocks over 2000–2025.
- A deterministic feature library of ~40 closed-form factors; no LLM used in V1.
- A Barra-lite cross-sectional risk model with 9 style factors and 24 GICS industry groups, estimated by √mcap-weighted WLS.
- A three-tier promotion gate (Promote / Hold / Retire) driven by Newey–West-adjusted IC t-stats, Benjamini–Hochberg FDR at $q = 0.10$, Harvey–Liu $|t| > 3.0$ hurdle, and Bailey–López de Prado Deflated Sharpe.
- A locked out-of-sample holdout of the final 24 months, plus a stricter sub-holdout of post-2023 months (per Huang & Fan 2026).
- FF3 / FF5 / FF6 risk-adjusted α with HAC t-statistics as a secondary validation, so a promoted factor must survive both residualization against Barra and abnormal-return regression against canonical academic models.
- A cost-aware long-short optimizer with explicit spread, Almgren impact, and tiered short-borrow terms.

What V2 and V3 add

- V2 introduces an LLM research engine over a 660-cell structured exploration grid (Wangpratham 2026) with priority-weighted cell selection, a RAG-grounded specification encoder, a Jaccard diversity gate on expressions, and a convergence-governed repair engine driven by a typed failure-mode taxonomy.

- V2 also adds Huang-Fan-style memory-guided state updates, Ledoit–Wolf shrinkage on the factor covariance, a two-state HMM for regime conditioning, and a LightGBM alternative to the linear ensemble.
- V3 extends to alternative data (10-K text, OptionMetrics), multi-agent critique, intraday execution simulation, and shadow paper trading through IBKR.

Foundational design principles

The five principles below are load-bearing. Every architectural and procedural choice elsewhere in this document derives from one or more of them, and any future extension that violates one of them should be treated as a regression rather than a feature.

- **Separation of concerns —** the LLM (V2) proposes ideas; deterministic, versioned code computes, scores, and rejects them. Evaluation never runs inside the language model. This keeps narrative plausibility from substituting for evidence: a candidate survives only if the §4 validation pipeline — HAC IC, BH-FDR, DSR, FF6 α, longer-horizon decay — says it survives.
- **Constrained search over free-form prompting —** the research engine searches inside an explicit grammar of variables, operators, horizons, and conditioning states, instantiated in V2 as the 660-cell exploration grid (Wangpratham 2026, §4.1.1). Search quality comes from good boundaries: a small, auditable token vocabulary + a six-rule deterministic validator + a Jaccard-0.70 diversity gate + a typed failure-mode catalogue. The generator is not asked to invent arbitrary new formulas; it is asked to fill defined grid cells.
- **Residual alpha before raw alpha —** no candidate is scored on raw predictive power alone. Every signal is first neutralized against the nine Barra-lite styles and twenty-four GICS industries (§4.4), then scored on specific-return IC. It is separately regressed on FF3/FF5/FF6 with HAC t-stats (§4.9). A candidate that only predicts because it inherits momentum, profitability, or size exposure is classified as a repackaging failure and rejected, not repackaged.
- **Structural alpha as a parallel research target —** the system searches not only for stand-alone signal formulas but for factor architectures: composite value constructions, momentum vetoes (junk exclusions, crash filters), tail-depth redesigns, refresh-timing choices, and contamination gates. These live in a dedicated V2 search axis alongside the single-signal axis, on the premise that much practical alpha lies in how known factors are packaged rather than in new primitives.
- **Auditability and replayability —** every candidate carries a full ledger: thesis string, formula AST, data dependencies and snapshot hashes, every metric computed against it, failure-mode classifications encountered, repair history with before/after diffs, and final status (Promoted / Held / Retired). Any row in that ledger can be replayed bit-for-bit from the code plus the PIT data snapshot. Nothing about a promoted signal is tacit.

2. Motivation: Alpha Decay as an Operational Rate

McLean and Pontiff (2016) document an average post-publication decay of roughly 5.6% per year for quantitative signals in U.S. equities; signals with high academic visibility decay at roughly twice that baseline. A live book of ~100 active signals must therefore retire and replace on the order of 5–10 signals per year merely to stay static. Wangpratham (2026) makes the same throughput argument at institutional scale.

Aegis Alpha Lab treats alpha decay as a throughput problem, not a research problem. V1 fixes a hand-specified library of ~40 deterministic factors so that the validation pipeline, risk engine, and portfolio optimizer are debugged and measured before any automated proposer is layered on. V2 then replaces the hand-specified library with an agentic proposer that targets the same validation pipeline, so that research throughput can scale without loosening the empirical discipline.

2.1 Research thesis

We decompose the forward return of stock *i* over period (*t*, *t*+1] into three additive components and state which we chase:

$$r_{i,t+1} = \alpha_{i,t} + \Sigma_k \beta_{i,k,t} \cdot f_{k,t+1} + \varepsilon_{i,t+1}$$

- Factor premia $\Sigma_k \beta \cdot f$ — real but crowded. Not the differentiator.
- Residual alpha α — signals that predict ε after β-exposures are stripped away. Primary research target of V1.
- Structural alpha — improvements to how known factors are packaged (tail depth, composite weighting, refresh timing, contamination filters) that raise their residual-alpha component. A V2 search module.

3. Notation

Symbol	Meaning
N_t	number of stocks in the eligible universe at date t (target ≈ 3,000)
T	number of rebalance dates in the sample (daily, ≈ 6,300 for 2000–2025)
F_t	information set (σ-algebra) generated by observable data up to t
r_{i,t+1}	arithmetic forward return of stock i over (t, t+1]
X_t ∈ R^{N_t × K}	matrix of factor exposures at t; K = 9 styles + 24 industries + 1 country
β_{i,k,t}	(i,k) entry of X_t; standardized cross-sectionally with √mcap weights
f_{t+1} ∈ R^K	cross-sectional WLS factor returns for period t+1

$\varepsilon_{i,t+1}$	stock-specific residual return
$s_{i,t}$	a candidate alpha signal for stock i at t, satisfying $s_{i,t} \in F_t$
$w_t \in R^{N_t}$	portfolio weights at t (dollar-neutral, so $1^T w_t = 0$)
Σ_t, Ω_t, D_t	total, factor-block, and specific-risk-block covariance at t

4. Mathematical Framework

4.1 Formal no-lookahead constraint

Following Huang and Fan (2026, Eq. 3.4), every candidate factor value is required to be measurable with respect to the information filtration at time t:

$$f_{i,t} \in F_t = \sigma(\{ X_{j,s} \}_{j \in N,\, s \leq t})$$

Every operator in the approved grammar — lags, rolling moments, cross-sectional ranks, Winsorize, z-score — preserves this measurability by construction. Cross-sectional transforms are applied per date using only same-day information; time-series transforms are applied per asset using only history up to and including t. Violations of this constraint terminate the candidate immediately; this is enforced by the validator, not by reviewer inspection.

4.2 Cross-sectional factor estimation (Barra-lite)

At each date t we run a weighted cross-sectional regression of realized forward returns on pre-known exposures. This is the USE3 / USE4 decomposition (Barra / MSCI).

$$\hat{f}_{t+1} = (X_t^T W_t X_t)^{-1} X_t^T W_t r_{t+1}$$

with diagonal weight matrix $W_t = \mathrm{diag}(\sqrt{\mathrm{mcap}_{i,t}})$. Exposures are winsorized at ±3 cross-sectional σ and then z-scored. Specific residuals are

$$\hat{\varepsilon}_{i,t+1} = r_{i,t+1} - \beta_{i,t}^T \hat{f}_{t+1}$$

4.3 Factor covariance and specific risk

$$\Sigma_t = X_t \Omega_t X_t^T + D_t$$

Ω_t is an EWMA covariance of $\{\hat{f}_s\}_{s \leq t}$ with half-life $\tau_\Omega = 90$ days. D_t is an EWMA of specific-return variance with half-life $\tau_D = 60$. V2 replaces Ω with a Ledoit–Wolf shrinkage estimator.

4.4 Residualized signal evaluation

Candidate s is first neutralized cross-sectionally against known exposures at each t:

$$s_t^{\perp} = s_t - X_t \, (X_t^{T} X_t)^{-1} X_t^{T} s_t$$

All IC-based evaluation operates on the residualized signal against specific residual returns.

4.5 IC, IC IR, and Newey–West adjusted t-statistic

$$IC_t^{\perp} = \rho_{Spearman}(s_t^{\perp} , \hat{\varepsilon}_{t+1})$$

$$IR_{IC} = \text{mean}(IC^{\perp}) / \text{std}(IC^{\perp}) \cdot \sqrt{252}$$

Because daily IC series are autocorrelated under overlapping labels, the t-statistic is reported with Newey and West (1987) HAC standard errors, following Huang and Fan (2026):

$$t_{IC} = \text{mean}(IC^{\perp}) / \sqrt{(\hat{V}_{NW} / T)}$$

4.6 Harvey–Liu multiple-testing hurdle

Harvey, Liu, and Zhu (2016) argue that the conventional $|t| > 2.0$ threshold is too permissive once the number of tested factors is taken into account, and recommend $|t| > 3.0$ as the new-factor hurdle in the post-factor-zoo era. V1 adopts this as a necessary condition for promotion:

$$| \, t_{IC} \, | > 3.0$$

This hurdle is combined with, not a substitute for, the FDR control and Deflated Sharpe gates below.

4.7 Benjamini–Hochberg FDR control

Within a batch of m candidates we compute the HAC-adjusted two-sided p-value of IR_{IC}. Sorting p-values ascending as $p_{(1)} \le \ldots \le p_{(m)}$, we reject every hypothesis with index at most

$$k^* = \max \{ i : p_{(i)} \le (i / m) \cdot q \}, \text{ with } q = 0.10$$

4.8 Deflated Sharpe Ratio (Bailey–López de Prado)

$$DSR = \Phi\left(\; [(SR - SR_0) \sqrt{(T-1)}] / \sqrt{(1 - \gamma_3 \cdot SR + (\gamma_4 - 1)/4 \cdot SR^2)} \; \right)$$

$$SR_0 = \sqrt{V[\hat{SR}]} \cdot [(1-\gamma) \cdot \Phi^{-1}(1 - 1/N) + \gamma \cdot \Phi^{-1}(1 - 1/(N \cdot e))]$$

N is the effective number of trials tracked in the research ledger. A candidate passes only if $DSR > 0.95$.

4.9 Incremental FF3 / FF5 / FF6 risk-adjusted α

Following Huang and Fan (2026, Exhibit 8), promoted long-short candidates are also regressed on Fama–French 3, 5, and 6-factor benchmarks with Newey–West HAC t-stats. Promotion requires:

$$\hat{\alpha}_{FF6} > 0 \quad \text{and} \quad t(\hat{\alpha}_{FF6}) > 2.5$$

This second-stage α test is specifically designed to detect signals that merely repackage canonical equity factors: if the candidate's edge disappears once momentum and profitability/investment are controlled, it fails.

4.10 Purged and embargoed walk-forward CV

For an h-day forward label, we purge any training observation whose label ends after the train boundary and embargo the first $e = 5$ trading days of the test fold (López de Prado 2018, ch. 7). V1 uses $h = 21$ and expanding year-long test folds.

4.11 Longer-horizon decay gate

Following Huang and Fan (2026, Exhibit 16), every promoted factor is also scored at holding horizons $H = 1, 2, 3, 5$, and 7 days. A candidate is retired if any of the following holds: (i) $H = 1$ t-stat ≥ 3.0 but $H = 3$ t-stat < 1.5 (symptom of microstructure noise); (ii) sign of IC flips between $H = 1$ and $H = 5$ (symptom of spurious short-horizon reversal).

5. Closed-Loop Research Workflow

We adopt the Huang–Fan (2026, Eq. 3.1) formalization of the research cycle:

$$C_k : H_k \text{ —Gen→ } F_k \text{ —Eval→ } M_k \text{ —Gate→ } D_k$$

Hypotheses H_k pass through generation, evaluation under the §4 protocol, and a gate that outputs a three-valued decision set D_k.

5.1 Three-tier promotion gate

The gate is rule-based, not discretionary. For every candidate f, the decision is

$$D(f) = \text{Promote} \quad \text{if} \quad t_{IC} \geq 3.0 \text{ and FDR-pass and DSR} > 0.95 \text{ and } t(\hat{\alpha}_{FF6}) > 2.5$$

$$D(f) = \text{Retire} \quad \text{if} \quad t_{IC} < 1.5 \text{ or DSR} < 0.50$$

$$D(f) = \text{Hold} \quad \text{otherwise}$$

Held candidates enter targeted robustness checks (regime sub-samples, alt-universe stress, longer-horizon decay). They are not re-scored against the original holdout.

5.2 Strict temporal and policy isolation

Per Huang and Fan (2026, Eq. 3.5), the promotion mapping Ψ is a function only of in-sample data:

$$D(f) = \Psi(\{ f_{i,t}, R_{i,t+1} \}_{t \in T\_IS})$$

Out-of-sample observations never enter Ψ, never re-rank candidates, and never update the next-round search policy. V1 defines T_IS = 2000–2022, T_OOS = 2023–2025, with 2024–2025 as a locked sub-holdout that is opened exactly once, after the rest of the system is frozen.

5.3 Memory-guided state update (V2 only)

For V2, the agent's search policy evolves per round via

$$S_{k+1} = \text{LLM}(S_k, E_k, L_k)$$

where E_k is the structured evaluation log and L_k is the catalogue of accepted, held, and retired candidates. This sequential, evidence-based update — as opposed to independent random re-draws — is the core mechanism by which V2 avoids redundant trials while still exploring diverse regions of the grammar.

6. V1 System Architecture

V1 compresses the original sketch into six modules. Each has a mathematical core, inputs, outputs, and an acceptance test.

Module	Core	Inputs	Outputs	Acceptance test
A. Data & PIT panel	corporate-action adjustment; PIT joins of CRSP + Compustat PIT; σ-algebra-tagged columns	CRSP daily, Compustat PIT, GICS PIT, SEC CIK map	stock-date Parquet panel, ≈ 15 M rows	reconstructed S&P 500 membership on any past date matches published index within 1 name
B. Research ledger	append-only SQLite: candidate_id, git SHA, config hash, metrics, gate status	candidate configs + metric dicts	queryable ledger; replay engine	every promoted factor replays bit-identical from ledger
C. Feature library	~40 deterministic closed-form factors (§8)	PIT panel	daily factor panel	reference factors (12–1 momentum, B/P) reproduce published IC to within 0.005
D. Barra-lite risk engine	cross-sectional √mcap-WLS (§4.2); EWMA cov (§4.3)	PIT panel, exposure definitions	X_t, f_t, $\hat{\varepsilon}_t$, Ω_t, D_t	mean $R^2 \geq 0.25$ across style factors; residual lag-1 $\lvert\rho\rvert \leq 0.05$
E. Validation & gate	purged CV + HAC t + BH-FDR + DSR + FF6 α + longer-horizon decay (§4.5–4.11, §5.1)	candidate signal, risk panel, labels	Promote / Hold / Retire + full metrics	false-positive rate on 1,000 synthetic null signals $\leq q = 0.10$
F. Portfolio & cost model	QP in §10 with cost model in §9	promoted signals, Σ_t, cost parameters	daily weights w_t, P&L, attribution	turnover matches QP dual; attribution residual ≤ 1 bp/day

7. Data Stack

Primary stack: CRSP + Compustat Point-in-Time via WRDS (Rice's Jones School subscription). This is the same provenance used by Gu, Kelly, and Xiu (2020) and by Huang and Fan (2026). Backup: Sharadar SF1/SEP at ~$300/month for survivorship-bias-free daily prices and PIT fundamentals. Free-stack fallback (Yahoo + SEC EDGAR XBRL) is documented as a V1 limitation and flagged in every metric.

Sample filters (Huang & Fan 2026, Exhibit 3): common shares only, NYSE/AMEX/NASDAQ listings, price ≥ $5, history ≥ 252 days. Target cross-section ≈ 3,000 stocks after filters, ≈ 16 M stock-day observations over 2000–2025. Forward returns winsorized cross-sectionally at the 1st and 99th percentiles.

8. V1 Feature Catalog

V1 enumerates ~40 closed-form factors. Each is specified by an explicit expression over the approved vocabulary. Cross-sectional z-score with 1%/99% winsorization is applied per date as in Huang and Fan (2026, Eq. 3.3).

8.1 Representative formulas

Value composite (equal-weighted z-score of four yield ratios):

$$V_{i,t} = ¼\ [\ z_t(B/P) + z_t(E/P) + z_t(S/P) + z_t(CF/P)\ ]$$

Momentum (12–1 month):

$$MOM_{i,t} = \log(\ P_{i,t-21} / P_{i,t-252}\ )$$

Gross profitability (Novy-Marx 2013):

$$Q_{i,t} = (\text{Revenue} - \text{COGS})_{i,t} / \text{Assets}_{i,t}$$

Amihud illiquidity (negated):

$$ILQ_{i,t} = -\ (1/21)\ \Sigma_{s=t-20}^{t}\ |r_{i,s}|\ /\ (\ P_{i,s} \cdot Vol_{i,s}\ )$$

Short-term reversal, low realized volatility, 60-day beta, and standard accruals / investment / growth factors round out the library. The full list with exact formulas lives in a versioned YAML in the repo.

9. Cost, Borrow, and Capacity Model

$$TC_{i,t} = s_{i,t} \cdot |\Delta w_{i,t}| + \eta \cdot (\ |\Delta w_{i,t}| \cdot GMV / ADV_{i,t}\ )^{3/2} + b_{i,t} \cdot \max(\ -w_{i,t}\ ,\ 0\ ) \cdot \Delta t$$

Corwin–Schultz daily half-spread floor (2 bp large, 5 bp mid), Almgren $\eta = 0.10$, tiered borrow (25 / 150 / 500 bp/yr by mcap decile). Huang and Fan (2026) use a flat 3 bp one-way cost as a simpler calibration; we report both under our full model and their 3-bp convention to enable apples-to-apples comparison with their 2.75/2.21 Sharpe benchmark.

Capacity cap: $|\ w_{i,t}\ | \le \min(\ w_{max}\ ,\ p_{ADV} \cdot ADV_{i,t} / GMV\ )$ with $w_{max} = 2\%$ and $p_{ADV} = 5\%$.

10. Portfolio Construction (V1)

$$\max_w\ \ w^T \hat{\alpha}_t - (\lambda/2)\ w^T\ \Sigma_t\ w - \kappa\ \|w - w_{t-1}\|_1 - \Sigma_i\ TC_{i,t}(w, w_{t-1})$$

subject to:

- $1^T w = 0$ — dollar-neutral
- $\beta\_mkt^T w = 0$ — beta-neutral to value-weighted CRSP market
- $|w\_i| \le 2\%$ — single-name concentration

- $|w\_i| \le 5\% \cdot ADV\_i / GMV$ — liquidity / capacity
- $|X\_s^T w| \le 0.25\ \sigma\_s$ — style-exposure budget (all 9 style factors)
- $|X\_I^T w| \le 0.50\ \sigma\_I$ — industry-exposure budget
- $GMV = \|w\|_1 = 1$ — gross leverage normalized to 1x

λ is chosen so the unconstrained solution has 8% ex-ante annualized vol; κ is tuned so realized turnover lands near 5× GMV/yr. OSQP / ECOS solves the 1,500–3,000-variable QP in < 1 s per date on a laptop.

11. Quantified Success Criteria

These thresholds are fixed before the locked sub-holdout is opened, and will not be relaxed afterward. Reference benchmarks from the literature are included so that V1 outcomes can be compared on a like-for-like basis.

Scope	Metric	V1 minimum	V1 target	Literature reference
Single factor	Residualized rank IC (mean)	0.02	0.04	Huang & Fan (2026), best factors 0.03–0.07
Single factor	IC IR (annualized)	0.50	0.75	—
Single factor	HAC t(IC)	3.0	5.0+	Harvey et al. (2016)
Single factor	Deflated Sharpe Ratio	0.95	0.99	Bailey & López de Prado (2014)
Single factor	FF6 α, HAC t	2.5	4.0+	Huang & Fan (2026), best single factors t ≈ 3–5
Composite book, gross	Annualized Sharpe	1.20	2.00	Huang & Fan (2026) report 2.75 gross
Composite book, net (3 bp)	Annualized Sharpe	0.80	1.50	Huang & Fan (2026) report 2.21 net
Composite book	Max drawdown	≤ 10%	≤ 6%	Huang & Fan (2026) 13.41% full window
Composite book	Turnover (GMV/yr)	≤ 8×	~5×	—
Composite book	Residual β to SPY	|β| ≤ 0.05	|β| ≤ 0.02	—

Failure in the locked holdout counts as V1 failing its research hypothesis. The engineering is still a success — but the decision then is to change the hypothesis in V2, not to retry with relaxed thresholds.

12. V1 Timeline (≈ 20 weeks)

Phase	Weeks	Deliverable

1. Data	1–5	WRDS pull, PIT joins, universe filters (common shares, price ≥ $5, 252-day history), unit tests for survivorship and restatement bias
2. Risk engine	6–8	Barra-lite estimation, EWMA covariance, diagnostic dashboard
3. Feature library	9–12	~40 closed-form factors, residualization, reference-factor unit tests
4. Validation stack	13–15	Purged walk-forward, HAC IC, BH-FDR, DSR, FF6 α, longer-horizon decay, synthetic null-signal false-positive test
5. Portfolio	16–18	QP optimizer, full cost + 3-bp-comparable runs, attribution
6. Lockbox	19–20	Freeze code, open 2024–2025 holdout, end-to-end run once, post-mortem

13. V2 Roadmap — Agentic Research Engine

V2 automates the proposer step while preserving every V1 validation and portfolio module. The target architecture is Wangpratham (2026)'s BrainAlpha, mapped onto our Barra-lite risk framework and long-only-or-short trading universe.

13.1 Structured 660-cell exploration grid

Alpha search is re-framed as exhaustible coverage over a Cartesian grid (Wangpratham 2026, §4.1.1):

$$\text{Grid} = 11\ (\text{signal families}) \times 5\ (\text{operator patterns}) \times 3\ (\text{horizons}) \times 4\ (\text{conditioning}) = 660\ \text{cells}$$

Signal families: value, quality, momentum, reversal, low-vol, liquidity, turnover, attention, accruals, flow, residual-return patterns. Operator patterns: linear, ratio, difference, rolling-moment, rank-spread. Horizons: short (1–5d), medium (21–63d), long (126–252d). Conditioning: none, high-vol regime, low-vol regime, dispersion-stressed regime.

Each cell has a lifecycle state in {empty, assigned, explored, exhausted}, enforcing breadth-first coverage. The planner picks the next cell by priority score (Wangpratham 2026, Eq. 1):

$$\text{Priority}(c, a) = 0.5 \cdot \text{Novelty}(c) + 0.3 \cdot \text{ExpectedYield}(c) + 0.2 \cdot \text{AgentAlignment}(c, a)$$

Novelty(c) $\in \{1.00, 0.50, 0.25, 0.00\}$ decreases monotonically with exploration status. ExpectedYield(c) is a family-level prior on pass rate, updated online by +0.10 after every promoted candidate in cell c. AgentAlignment(c, a) $\in \{0.3, 0.5, 0.7, 1.0\}$ captures the fit between cell c and the specialty of agent a.

13.2 RAG-grounded specification

A vector index over the live feature vocabulary (≈ 10,000 allowed field × operator tokens) retrieves the top-k relevant primitives for each thesis. The generator is constrained to sample only from retrieved tokens plus the fixed operator alphabet. A six-rule deterministic validator rejects any candidate that fails:

non-empty primary fields, lookback ∈ [1, 120], recognized neutralization strategy, non-empty operator family, non-empty direction, hypothesis string ≥ 20 characters. Validation is zero-cost — no LLM call, no simulation budget consumed.

13.3 Jaccard diversity gate

A new expression is admitted only if its operator-token Jaccard similarity against every accepted expression is below 0.70 (Wangpratham 2026, Eq. 2):

$$\max_{e \in A} \; |\, \mathrm{ops}(e_{new}) \cap \mathrm{ops}(e) \,| \,/\, |\, \mathrm{ops}(e_{new}) \cup \mathrm{ops}(e) \,| \; < \; 0.70$$

This prevents lexical near-duplicates from crowding the library, the single most common failure mode of naive LLM factor-mining.

13.4 Typed failure-mode taxonomy

Evaluation failures are classified into six typed modes (Wangpratham 2026, Table 3), each mapped to an ordered catalogue of targeted repairs:

FM	Failure mode	Diagnostic	Primary repair actions
FM-1	Low signal quality	HAC t(IC) < 3.0, flat PnL	style neutralization, industry neutralization, longer lookback, winsorize
FM-2	Excessive turnover	turnover > cap	decay_linear smoothing, longer lookback
FM-3	High drawdown	max DD > cap	volatility conditioning, winsorize tails
FM-4	Signal reversal	IC sign opposite to thesis	negation, inversion
FM-5	Low coverage	long/short count < min	pasteurize, relax universe filter
FM-6	Low turnover / stale	turnover < floor	shorten lookback, add cross-sectional rank

13.5 Convergence-governed repair engine

Before any repair iteration, three hard-stop conditions trigger immediate termination (Wangpratham 2026, §4.4.1):

- I1 — self-correlation fail: structural duplication of an already-promoted signal; expression-level repair cannot fix idea-level duplication.
- I2 — reversed complex signal: IC ≤ 0 and AST complexity > 20 nodes; negation cannot propagate cleanly through deep nesting.
- I3 — multi-failure: four or more failing checks simultaneously; no single-action repair addresses four independent deficiencies.

Five stop conditions end the loop once it begins: (i) all gates pass; (ii) max iterations reached; (iii) per-iteration Sharpe improvement < $\delta = 0.02$ for three consecutive rounds; (iv) gap-percentage widens rather than shrinks (divergence); (v) repair catalogue exhausted. Together, I1–I3 and the five stop conditions cap the number of repair rounds per candidate at a small constant, eliminating the search-budget-abuse risk of unbounded retry loops.

13.6 Temperature protocol

Generator calls use $T = 0.7$ for exploration. Repair / rewrite calls use $T = 0.3$, so revisions are targeted edits of a failing expression rather than fresh attempts from scratch (Wangpratham 2026, §4.4.3).

13.7 Ensemble: linear plus LightGBM

V1's transparent linear ensemble remains the auditable baseline. V2 adds a LightGBM non-linear aggregator (Huang and Fan 2026, §5.2) trained over the promoted-factor matrix. Both ensembles run in parallel; promotion of the LightGBM ensemble over the linear requires a Diebold–Mariano test on out-of-sample Sharpe to clear 5% significance. This prevents spurious gains from model-class flexibility alone.

13.8 Regime conditioning, covariance shrinkage, crowding

- Two-state HMM on (realized vol, CRSP dispersion, credit-spread change). Regime labels use filtered (one-sided) posterior only.
- Ledoit–Wolf shrinkage on Ω with data-driven shrinkage intensity δ^*.
- Eigen-adjustment (Menchero, Poduri & Zhang 2011) to correct the upward bias of the sample minimum-variance portfolio.
- Dynamic borrow fees from a stock-loan feed, replacing mcap-decile tiers.
- Crowding proxy (hedge-fund 13F overlap, short-interest ratio) added as a penalty term in the QP.

14. V3 Roadmap — Alternative Data, Multi-Agent, Execution

14.1 Alternative data

- 10-K / 10-Q text via SEC EDGAR: sentence-embedding similarity signals (Cohen, Malloy & Nguyen 2020).
- Option-implied skew, risk-reversal, and IV spread from OptionMetrics IvyDB.
- Short-interest changes from NYSE / Nasdaq settlement reports.
- I/B/E/S consensus revisions if WRDS includes it.

14.2 Multi-agent critique

A Proposer agent specialized on a single signal family, a Skeptic agent whose job is to attribute every candidate to an existing Barra factor or already-in-library anomaly, and an Economist agent that must supply a falsifiable economic channel. Human reviewer approval is required for any candidate promoted out of V3 search.

14.3 Intraday execution simulation

TAQ-based simulator honoring visible book depth and converting daily QP weights to a VWAP or Almgren–Chriss trajectory. Replaces the static cost model with realized slippage distributions rather than point estimates.

14.4 Shadow paper trading

End-of-day weights routed to an IBKR paper account via their API, with position reconciliation and live cost attribution versus the simulated cost model. Runs for at least 6 months before any live-capital discussion.

15. Mathematical and Computational Feasibility

15.1 Data volume

~3,000 names × ~6,300 trading days = 1.9×10^7 rows × ~60 columns. Parquet storage ≈ 6 GB, DuckDB indexed ≤ 20 GB. A 32 GB / 1 TB SSD laptop is sufficient.

15.2 Risk-engine compute

Cross-sectional WLS: $K \approx 34$, $N \approx 3{,}000$, a 34×34 normal-equations inversion per date. One full sweep of 6,300 dates runs in under 10 s. EWMA covariance is $O(K^2)$ per date — negligible.

15.3 Portfolio optimization

Daily QP: ~3,000 variables, ~60 linear constraints, factor-structured covariance so the solver never materializes the full $N \times N$ Σ. OSQP < 1 s per date; 6,300 dates serially in under two hours, minutes in parallel.

15.4 Statistical power

With mean residualized IC 0.03 and $\sigma \approx 0.06$, HAC standard error of mean IC over $T = 252 \times 18 \approx 4{,}500$ daily obs is ≈ 0.00089. Population $t \approx 33$ — the bottleneck is multiple-testing control, which §4.6–4.9 addresses directly.

15.5 LLM budget (V2)

Wangpratham (2026) reports BrainAlpha saturating the 660-cell grid in the low-thousands of LLM calls. At current API pricing, that is in the range of low-hundreds of dollars per full grid sweep — negligible relative to the marginal value of a promoted signal.

15.6 Tooling: Qlib as a selective accelerator

Microsoft Qlib (Yang et al. 2020) is an open-source quantitative research platform that overlaps our design on three layers and conflicts with it on two. It is worth adopting selectively rather than wholesale.

Where Qlib accelerates Aegis Alpha Lab

- **Data layer —** Qlib's binary columnar store and its DataHandler / Alpha158-style expression engine are a near-perfect match for the point-in-time panel described in §7. We can ingest the WRDS CRSP + Compustat PIT Parquet into Qlib's binary format for 10–50× faster feature materialization versus pandas groupby, without changing the provenance guarantees.
- **Expression DSL as scaffold for the V2 grammar —** Qlib's symbolic operator set (Ref, Rank, Mean, Std, Corr, Cov, Delta, Sum, IdxMax, Slope, Log, Sign, Abs, WMA, EMA, …) is a published, tested superset of the operator alphabet needed by our Wangpratham-style 660-cell grid (§13.1). Adopting it directly — rather than rewriting — gives the V2 generator a proven, ruleset-level-safe token vocabulary and makes the six-rule deterministic validator (§13.2) trivial to implement.
- **Workflow recorder —** Qlib's qlib.workflow.R recorder stores each experiment's config, dataset hash, model parameters, metrics, and artifacts under an MLflow-compatible tree. This is a direct fit for principle 5 (Auditability and replayability): the research ledger becomes a set of Recorder IDs, and bit-exact replay is a one-liner.
- **Rolling training utilities —** qlib.contrib.rolling implements expanding-window and rolling-window retraining that maps cleanly onto our purged-embargoed walk-forward CV (§4.10). The purge and embargo logic still needs to be added — Qlib does not do this out of the box — but the rolling scaffolding is correct.
- **Model zoo for the V2 LightGBM ensemble —** qlib.contrib.model.gbdt provides a preset LightGBM pipeline with CV, feature importance, and early stopping already wired. It is a drop-in comparison baseline for principle 1 (Separation of concerns): the language model proposes, Qlib's deterministic LightGBM aggregates.

Where Qlib is the wrong abstraction and must be replaced

- **Risk decomposition —** Qlib has no Barra-style cross-sectional factor model. Our §4.2 WLS estimation of style + industry factor returns, the §4.3 factor-structured covariance, and the §4.4 residualized-signal scoring must all be written against Qlib's panel — not delegated to it. Residual alpha (principle 3) is strictly out of Qlib's scope.

- **Multiple-testing discipline —** Qlib reports per-strategy ICIR, Rank-ICIR, and annualized Sharpe, but does not implement Newey–West HAC IC t-stats (§4.5), Benjamini–Hochberg FDR (§4.7), Bailey–López de Prado Deflated Sharpe (§4.8), or the Harvey–Liu $|t| > 3.0$ hurdle (§4.6). All four must be added on top.
- **Cost model —** Qlib's default TopkDropoutStrategy cost model is a flat per-side bp charge. Our §9 Corwin–Schultz spread + Almgren $\eta|\Delta w|^{3/2}$ impact + tiered borrow model is strictly richer and must be added as a custom strategy class.
- **Portfolio optimization —** Qlib's built-in strategies are rank-based long-only (TopkDropout). The §10 cost-aware QP with dollar/beta neutrality, style/industry exposure budgets, and ADV capacity caps is outside Qlib; we retain OSQP and plug the resulting weights back into Qlib's backtest engine for attribution only.

Recommended integration pattern

Use Qlib as infrastructure, not as framework. Concretely, adopt Qlib for (data loader, expression DSL, recorder, rolling scaffolding, LightGBM baseline) and write custom code for (Barra-lite WLS risk engine, HAC/FDR/DSR/FF6 validation stack, cost model, QP optimizer, promotion gate). The §6 module boundaries already correspond to this split: Module A (Data & PIT panel) and parts of Module C (Feature Catalog) sit inside Qlib; Modules B (Risk), D (Validation), E (Portfolio) sit outside. This preserves the separation-of-concerns principle: Qlib owns bulk storage and feature materialization, but never owns the decision to promote a signal.

Practical checklist: (1) pin Qlib to a specific commit in requirements.lock; (2) wrap every Qlib API call in a thin adapter module so Qlib can be swapped without touching validation or portfolio code; (3) mirror every Qlib Recorder artifact into the independent research ledger so replayability does not depend on MLflow being online; (4) disable Qlib's default strategy backtest and drive backtest from our own QP weights only. This contains Qlib's opinionation behind a seam and keeps the five principles enforceable at the seam.

16. References

- Bailey, D. H., and López de Prado, M. (2014). The Deflated Sharpe Ratio. Journal of Portfolio Management 40(5), 94–107.
- Benjamini, Y., and Hochberg, Y. (1995). Controlling the false discovery rate. JRSS-B 57, 289–300.
- Carhart, M. (1997). On persistence in mutual fund performance. Journal of Finance 52, 57–82.
- Corwin, S. A., and Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. Journal of Finance 67, 719–760.
- Fama, E. F., and French, K. R. (1993, 2015). Common risk factors / A five-factor asset pricing model. Journal of Financial Economics.

- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. Review of Financial Studies 33, 2223–2273.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). … and the cross-section of expected returns. Review of Financial Studies 29, 5–68.
- Huang, A. Y., and Fan, Z. (2026). Beyond Prompting: An Autonomous Framework for Systematic Factor Investing via Agentic AI. arXiv:2603.14288v2.
- Jegadeesh, N., and Titman, S. (1993). Returns to buying winners and selling losers. Journal of Finance 48, 65–91.
- Ledoit, O., and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. JMVA 88, 365–411.
- López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley. (Purged k-fold: ch. 7; False-Strategy Theorem: ch. 8.)
- McLean, R. D., and Pontiff, J. (2016). Does academic research destroy stock return predictability? Journal of Finance 71, 5–32.
- Menchero, J., Orr, D. J., and Wang, J. (2011). The Barra US Equity Model (USE4). MSCI Research Insight.
- Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity- and autocorrelation-consistent covariance matrix. Econometrica 55, 703–708.
- Novy-Marx, R. (2013). The other side of value: the gross profitability premium. Journal of Financial Economics 108, 1–28.
- Wangpratham, N. (2026). BrainAlpha: An Autonomous Multi-Agent System for Quantitative Alpha Discovery. QuantCorner Laboratory working paper, SSRN 6313578.